

AHMAD TOBASEI

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EDUCATION

University of Tennessee

B.S. Computer Science; Minor in Machine Learning, GPA: 3.7

Knoxville, TN

Aug. 2022 – May 2026

EXPERIENCE

Remote Sensing Research Assistant

SEER Lab, University of Tennessee

Sep. 2024 – May 2026

Knoxville, TN

- Automated mosaicking and band harmonization pipelines in Python using SLURM managing 50–100+ GB datasets and chained job dependencies
- Handled recurring out of memory and disk quota failures by repartitioning jobs and changing how intermediate files were cleaned up, allowing large file operations to complete end to end
- Applied Random Forest models for land cover classification on field survey data, resampling iteratively to improve accuracy from 60.3% to 81.2%

Developer

Hack4Impact

Sep. 2023 – May 2024

Knoxville, TN

- Built the dashboard list component for the organization's applicant tracking system, working in scrum sprints on a 6 person team
- Implemented frontend and backend features using JavaScript and HTML/CSS and integrated database functionality using MongoDB

PROJECTS

Fake Job Posting Predictor | *Python, LLMs, RAG, Chrome Extension*

- Created a Retrieval Augmented Generation (RAG) system to classify online job postings as real or fraudulent and provide explanations for each result.
- Integrated this system into a Chrome extension allowing for in browser use

Unix Shell Implementation | *C, Linux*

- Created a shell based on Bash supporting Unix command parsing, multiprocessing, I/O redirection, and signal handling

Skincare Recommender | *Python, React, Node.js, Firebase*

- Developed a skincare recommendation platform scoring products on user preferences such as skin type, concerns, and price filters
- Implemented a Python based selection algorithm connected to a React frontend and Node.js backend, with Firebase handling authentication and storage

Automated Network Compliance & Configuration Manager | *Python, JSON, Networking*

- Built a system to retrieve and compare network device configurations against baseline compliance policies
- Detected configuration errors, generated remediation actions, and designed parsing logic and rule sets

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL

Frameworks: React, Node.js, Flask

Developer Tools: Git, GitHub, Docker, Linux, SLURM, MongoDB, QGIS

Libraries: scikit-learn, NumPy, pandas, geopandas

RELEVANT COURSEWORK

Data Structures and Algorithms, Systems Programming, Software Engineering, Machine Learning, Deep Learning, Advanced LLMs, Cybersecurity